

Artificial Intelligence, Accountability, and Ethical Governance in Modern Digital Systems

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**“THERE IS A GROWING
RECOGNITION THAT THE ERA OF
SELF-REGULATION IN AI IS ENDING.”**

Abstract

Artificial intelligence (AI) has rapidly evolved from a specialized technological tool into a foundational component of decision-making across healthcare, finance, education, criminal justice, and public administration. While AI offers significant opportunities to improve efficiency, innovation, and economic growth, its widespread adoption has also introduced complex legal and ethical challenges involving algorithmic bias, privacy, transparency, accountability, and the protection of fundamental rights. Existing governance approaches remain fragmented and often rely on voluntary ethical principles that lack meaningful enforcement mechanisms. This article argues that ethical principles alone are insufficient to govern increasingly autonomous AI systems. Instead, effective AI governance requires legally enforceable accountability mechanisms that establish clear responsibilities for AI developers, deployers, and users while ensuring transparency, explainability, meaningful human oversight, and appropriate legal remedies for individuals affected by automated decision-making. To support this argument, the article evaluates emerging governance frameworks, including UNESCO's Recommendation on the Ethics of Artificial Intelligence, the European Union's AI Act, the United States' NIST AI Risk Management Framework, and the OECD AI Principles, examining how different jurisdictions balance technological innovation with the protection of human rights. Building upon these frameworks, this article proposes a governance approach centered on mandatory transparency, independent algorithmic auditing, human oversight in high-risk applications, and clearly defined legal accountability throughout the AI lifecycle. It concludes that the long-term legitimacy of artificial intelligence will be determined not by the sophistication of the technology itself, but by the willingness of legal institutions to transform ethical principles into enforceable legal obligations that promote public trust, protect fundamental rights, and support responsible innovation.

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I. INTRODUCTION

Artificial intelligence has become an integral component of contemporary decision-making, influencing outcomes in healthcare, finance, employment, criminal justice, and public administration. It diagnoses diseases, determines creditworthiness, influences hiring decisions, and increasingly shapes the outcomes of judicial proceedings. [1] Yet the legal and regulatory frameworks governing these systems remain fragmented, voluntary, and largely aspirational. The gap between what AI can do and what the law requires of those who deploy it has grown into one of the defining governance challenges of the twenty-first century.

The proliferation of voluntary ethics frameworks developed by technology companies, governments, and international organizations has undoubtedly advanced responsible AI discourse. However, these frameworks generally lack binding legal obligations, standardized enforcement mechanisms, and effective remedies for individuals harmed by automated decision-making.

The stakes are substantial. When an AI system incorrectly denies a loan application, misidentifies a criminal suspect, or recommends a harmful medical treatment, the affected individual rarely has meaningful recourse. The developer may disclaim responsibility, the deploying company may point to the algorithm, and regulators may lack the authority or expertise to intervene. [31] Although the accountability gap [2] has technical dimensions, its most significant implications are legal, as existing regulatory systems often fail to allocate responsibility when

algorithm system cause harm. Several jurisdictions are now moving toward more structured and enforceable governance. The European Union's Artificial Intelligence Act, adopted in 2024, [3] represents the most comprehensive regulatory framework to date, classifying AI systems by risk level and imposing obligations of transparency, human oversight, and conformity assessment on high-risk applications. In the United States, the National Institute of Standards and Technology released its AI Risk Management Framework in 2023, [4] providing voluntary guidance for organizations developing and deploying AI systems across sectors. In New York City, Local Law 144 [5] introduced mandatory bias audits for automated employment decision tools, marking one of the first binding municipal AI regulations in the country. At the international level, UNESCO's Recommendation on the Ethics of Artificial Intelligence further reflects a growing global consensus that AI governance should be grounded in human rights, transparency, accountability, fairness, and meaningful human oversight. [22] Although the Recommendation is not legally binding, it provides an internationally recognized ethical framework that increasingly influences national AI policies and regulatory initiatives. These developments signal a growing recognition that the era of self-regulation in AI is ending.

This article proceeds in four parts. Part II examines the ethical challenges that current AI systems present, including algorithmic bias, opacity, privacy violations, and the erosion of human oversight, drawing on recent legal disputes and documented harms to ground the analysis in concrete reality rather than theoretical risk. Part III surveys the emerging landscape of AI governance across major jurisdictions, comparing the EU's risk-based regulatory model, the U.S. sector-specific approach, and the principles-based frameworks adopted by international bodies including UNESCO and the OECD. Part IV develops an enforceable AI governance framework by examining

mandatory transparency obligations, human oversight in high-risk applications, independent algorithmic auditing, and legal liability for algorithmic harm. The article concludes that the legitimacy of AI in public and private life depends on society's willingness to subject it to the same rule of law that governs every other consequential technology.

II. ETHICAL CHALLENGES AND LEGAL RISKS IN AI DEPLOYMENT

The challenges posed by artificial intelligence extend well beyond theoretical ethical debates. As AI systems increasingly influence decisions involving liberty, employment, healthcare, finance, and public safety, failures in their design or deployment can produce tangible legal and societal harms. From biased hiring tools and flawed criminal risk assessments to privacy violations and misdiagnoses in clinical settings, what emerges is not a theoretical warning but a documented pattern of harm that existing legal frameworks have consistently failed to address.

These incidents reveal systemic shortcomings in accountability, oversight, and regulatory enforcement. Voluntary ethical guidelines, while valuable in articulating aspirational principles, have proven insufficient to prevent or remedy the harms documented across multiple sectors. This Part examines five domains in which AI deployment has generated well-substantiated harms, demonstrating why enforceable governance mechanisms not merely ethical commitments are necessary to safeguard fundamental rights and maintain public trust. These examples establish the factual and legal foundation for the governance framework proposed in Part IV.

A. Algorithmic Bias in Employment

Among the earliest and most consequential demonstrations of AI bias in a corporate setting was Amazon's internal recruitment tool, developed

between 2014 and 2018 to automate the screening of job applicants. The system, trained on historical hiring data, learned to penalize resumes containing words associated with women, including the names of women's colleges and systematically downgraded female candidates for technical roles. [6] When Amazon engineers discovered the bias, the company abandoned the tool entirely [7] rather than attempt to correct it. The episode demonstrates that algorithmic decision-making, despite objective and impartial outcomes, can reflect the assumptions and biases embedded in the data on which such systems are trained. More broadly, AI systems trained on historical datasets can reproduce and amplify existing social biases, raising significant legal concerns regarding discrimination, equal employment opportunity, and organizational accountability. [32] [40] Importantly, the absence of discriminatory intent does not eliminate legal responsibility when automated systems produce discriminatory outcomes. This distinction is particularly significant in employment law, where liability may arise from discriminatory effects even when there is no evidence of intentional discrimination.

New York City's response to this class of problem is instructive. Local Law 144, enacted in 2021 and enforced beginning in 2023, [5] requires employers using automated employment decision tools to conduct and publish annual independent bias audits before deploying such systems. The law also mandates disclosure to job candidates that an automated tool is being used. While limited in scope, Local Law 144 represents a model of enforceable algorithmic accountability one that moves beyond voluntary commitments and imposes binding legal obligations on organizations deploying AI systems. More importantly, Local Law 144 reflects a broader shift in AI governance from reactive compliance toward preventive regulation. Rather than waiting for discriminatory outcomes to emerge through litigation, the law requires organizations to evaluate algorithmic systems before deployment through independent

bias audits and transparency obligations. [33] This approach recognizes that accountability should be embedded throughout the AI lifecycle rather than imposed only after harm has occurred. As similar regulations emerge across other jurisdictions, algorithmic auditing may become a foundational component of responsible AI governance. The Amazon case and New York City's legislative response illustrate a broader shift in AI governance from voluntary ethical commitments toward enforceable legal accountability. Together, they demonstrate that algorithmic bias in machine learning systems presents a broader governance challenge requiring clear regulatory standards, institutional oversight, and meaningful accountability. As AI increasingly influences employment and other high-stakes decisions, organizations must be held responsible for ensuring that automated systems comply with principles of fairness, transparency, and non-discrimination.

B. Opacity and Due Process in Criminal Justice

Perhaps no deployment of AI raises more urgent legal concerns than its use in criminal sentencing. COMPAS (Correctional Offender Management Profiling for Alternative Sanctions) is a proprietary risk assessment tool used by courts in several U.S. states to predict the likelihood that a defendant will reoffend. Scores generated by COMPAS have been used to inform decisions about bail, sentencing, and parole decisions with profound consequences for individual liberty. [34] Because these decisions directly affect an individual's constitutional rights and personal liberty, the use of AI-assisted risk assessments raises fundamental questions regarding procedural fairness, judicial discretion, and the extent to which automated recommendations should influence legal decision-making.

A 2016 investigation by ProPublica found that COMPAS scores were racially biased: black defendants were nearly twice as likely as white defendants to be falsely flagged as high risk for future offending. [8] The

tool's developer, Northpointe (now Equivant), disputed the findings, arguing that COMPAS was equally predictive across racial groups and that differences in the error rates did not, by themselves, demonstrate racial bias. [35] The dispute nevertheless highlighted a broader structural problem: because COMPAS is proprietary software, defendants and their attorneys have no meaningful ability to examine how the system generates its scores or to challenge its accuracy. [36] In *State v. Loomis*, the Wisconsin Supreme Court upheld the use of COMPAS scores in sentencing while acknowledging that the defendant had no ability to access or contest the algorithm, [9] a troubling concession that due process concerns were subordinated to efficiency. The COMPAS case illustrates what legal scholars call the "black box" problem in AI governance: [2] when consequential decisions are made by opaque algorithms, the right to a fair hearing becomes functionally meaningless. Transparency and explainability are fundamental safeguards in contexts where AI influences judicial decision-making, supporting procedural fairness, protecting due process, and maintaining public confidence in the administration of justice.

The use of COMPAS therefore reinforces a central argument of this article: when AI systems participate in high-stakes public decision-making, voluntary ethical principles alone are insufficient to protect individual rights. Legally enforceable requirements for transparency, explainability, human oversight, and judicial accountability are necessary to ensure that technological innovation remains consistent with the principles of due process, equality before the law, and the rule of law itself.

C. Privacy Violations and Facial Recognition

Clearview AI, a facial recognition company, scraped billions of images from social media platforms without user consent to build a database sold to law enforcement agencies across the U.S. and internationally.

[10] The company's technology enables law enforcement to identify individuals from a single photograph, a capability with obvious investigative value and equally obvious potential for abuse.

Regulators in multiple jurisdictions found Clearview's practices unlawful. The UK Information Commissioner's Office issued a £7.5 million fine in 2022, finding that the company had violated data protection law by collecting images without a lawful basis. [11] Similar findings followed in Australia, Canada, France, and Italy. The United States, where no comprehensive federal privacy law existed at the time, enforcement was limited to state-level actions under the Illinois Biometric Information Privacy Act. [12]

The Clearview AI episode demonstrates that AI-driven privacy violations can occur at a scale and speed that exceed the capacity of many existing legal frameworks, exposing significant gaps in privacy regulation, regulatory enforcement, and cross-border governance. It also underscores the consequences of regulatory fragmentation: when privacy protection depends on the jurisdiction in which an individual happens to reside, the protection is structurally unequal.

Clearview AI therefore illustrates a central argument of this article: effective AI governance cannot depend on fragmented regulatory approaches or voluntary corporate commitments. Meaningful protection of privacy requires harmonized legal standards, enforceable consent requirements, robust regulatory oversight, and clear accountability for organizations that develop and deploy AI-powered surveillance technologies. Without such safeguards, advances in facial recognition risk undermining fundamental rights while eroding public trust in the responsible use of artificial intelligence.

D. Accountability in Legal and Commercial Contexts

In 2023, two U.S. attorneys submitted a court brief containing citations

to six legal cases that did not exist. [13] The citations had been generated by ChatGPT, which produced plausible sounding but entirely fabricated case names, docket numbers, and judicial quotations. When the court discovered the error, the attorneys were fined and publicly sanctioned. The incident prompted federal judges across the country to issue standing orders requiring disclosure of AI use in legal filings.

A related but distinct accountability question arose in the case *Moffatt v. Air Canada* (2024), in which the Civil Resolution Tribunal of British Columbia held Air Canada liable for inaccurate information provided by its customer service chatbot. [14] The company had argued that the chatbot was a separate legal entity whose representations did not bind the airline, an argument the Tribunal rejected, holding that a company cannot disclaim responsibility for information disseminated through its own automated systems. The ruling established an important precedent: organizations deploying AI-powered customer interfaces bear legal responsibility for the outputs those systems produce.

Together, these cases demonstrate that the accountability gap in AI has moved well beyond theory, producing concrete legal disputes, and courts are beginning to fill the regulatory void that legislatures have left. These disputes also illustrate an important shift in AI governance. Rather than treating algorithmic failures as isolated technological errors, courts are increasingly evaluating them through established principles of negligence, agency, professional responsibility, consumer protection, and organizational liability. This judicial approach suggests that while comprehensive AI legislation continues to evolve, existing legal doctrines remain capable of providing meaningful accountability when organizations fail to exercise appropriate oversight over automated systems.

E. AI in Healthcare: Diagnostic Bias and Liability

The deployment of AI in healthcare settings presents the accountability

gap in a domain where its consequences are uniquely difficult to remedy. AI diagnostic tools are now used to detect cancer in radiological images, predict patient deterioration in intensive care units, and guide treatment recommendations for conditions ranging from sepsis to mental health disorders. The potential benefits are significant; studies suggest that AI diagnostic tools can match or exceed the accuracy of specialist physicians in specific tasks. The risks, however, are equally significant and less frequently acknowledged.

Research published in leading medical journals has documented systematic racial and ethnic bias in AI diagnostic systems. A study published in *Science* found that AI models trained predominantly on data from white patients performed substantially less accurately on patients of other racial and ethnic backgrounds, a disparity with direct implications for health equity. [15] In dermatology, AI skin cancer detection tools [16] trained on datasets lacking sufficient representation of darker skin tones have demonstrated lower diagnostic accuracy for Black patients, a finding that has prompted calls for mandatory diversity requirements in medical AI training data.

Liability questions in AI assisted healthcare remain largely unresolved. When an AI system recommends a treatment that harms a patient, the question of who bears legal responsibility, the software developer, the hospital, the attending physician, or some combination has no settled answer in U.S. law. The Food and Drug Administration has begun to develop frameworks for regulating AI-enabled medical devices, but the pace of regulatory development has lagged behind the pace of clinical deployment. [17] The result is a landscape in which patients may be harmed by AI systems without clear legal pathway to remedy.

The healthcare context makes the case for enforceable AI governance with particular force. When the stakes are a patient's life or health,

voluntary ethical commitments by technology companies are not a sufficient substitute for legal accountability.

The healthcare sector therefore exemplifies the central argument of this article. While ethical principles such as fairness, transparency, and beneficence remain essential, they must be supported by enforceable legal obligations that clearly define responsibility, establish appropriate regulatory oversight, and provide effective remedies when AI systems cause harm. As AI becomes increasingly integrated into clinical practice, maintaining public trust will depend not only on technological accuracy but also on the strength of the legal frameworks governing its development and deployment.

Collectively, the cases examined throughout this Part demonstrate that the most pressing challenges posed by artificial intelligence are rooted not in what the technology cannot do, but in what the law has failed to require. They reveal a consistent pattern in which rapid innovation has outpaced the legal frameworks necessary to ensure transparency, accountability, and effective protection of fundamental rights.

III. THE EMERGING LANDSCAPE OF AI GOVERNANCE

The recognition that AI requires regulatory attention has produced a diverse range of regulatory and governance responses across jurisdictions and international institutions. These responses reflect fundamentally different assumptions about the appropriate role of law in managing technological risk; assumptions rooted in distinct legal traditions, political economies, and relationships between the state and the private sector. This Part surveys major governance frameworks, identifies their respective strengths and limitations, and argues that a risk based, legally enforceable governance model provides the most effective foundation for an AI accountability framework capable of protecting fundamental rights while supporting responsible innovation.

Although these frameworks share common objectives including promoting trustworthy AI, protecting fundamental rights, and encouraging responsible innovation, they differ significantly in their legal force, enforcement mechanisms, and regulatory scope. Some rely primarily on voluntary guidance and industry self-governance, while others establish binding legal obligations backed by regulatory oversight and sanctions. Examining these differences is essential for understanding how legal systems can balance technological innovation with public accountability and why certain governance models are better equipped to address the accountability gap identified in Part II.

A. The European Union: A Risk-Based Regulatory Model

The European Union's Artificial Intelligence Act, which entered into force in August 2024 represents the most comprehensive binding legal framework for AI governance yet enacted anywhere in the world. [18] The Act adopts a risk-based architecture that classifies AI systems into four categories: unacceptable risk, limited risk, high risk, minimal risk and impose legal obligations proportionate to the potential harm each category presents. Unlike many earlier AI governance initiatives that relied primarily on voluntary ethical principles, the EU AI Act establishes legally enforceable obligations supported by regulatory oversight and financial penalties. In doing so, it reflects a preventive approach to AI governance that seeks to reduce foreseeable risks before they result in legal disputes or societal harm.

AI systems classified as presenting unacceptable risk are prohibited outright. These include systems that deploy subliminal manipulation to influence behaviour, exploit vulnerabilities of specific groups, enable real time biometric surveillance in public spaces by law enforcement (with limited exceptions), and create social scoring systems for general purposes. These restrictions reflect the view that certain uses of AI conflict with human dignity and the principles of democratic governance, regardless of how advanced or commercially valuable the technology may be.

High risk AI systems used in critical infrastructure, education, employment, essential services, law enforcement, migration, and the administration of justice are subject to the most demanding obligations under the Act. Developers and deployers of high-risk systems must implement risk management systems, ensure the quality and representativeness of training data, maintain technical documentation, enable human oversight, achieve accuracy and robustness standards, and register their systems in a publicly accessible EU database. High-risk systems require independent conformity assessment prior to deployment.

For foundation models and general-purpose AI systems a category added in response to the rapid development of large language models, the Act imposes transparency obligations, requires documentation of training data, and mandates compliance with EU copyright law. Developers of the most powerful foundation models, defined by computational thresholds, face additional obligations including adversarial testing and incident reporting.

The penalties for non-compliance are substantial. Violations of the prohibited practices provisions may attract fines of up to thirty-five million euros or seven percent of global annual turnover, [19] whichever is higher. The Act also establishes a European AI Office within the European Commission to coordinate enforcement and develop technical standards.

The EU AI Act is not without limitations. Critics have raised concerns that its compliance requirements may disadvantage smaller developers relative to large technology companies with dedicated legal and regulatory teams. Others have questioned whether the risk classification system is sufficiently dynamic to keep pace with rapidly evolving AI capabilities. Nevertheless, as a demonstration that

comprehensive, binding AI regulation is legally and institutionally achievable, the EU AI Act is a landmark development that has already begun to influence regulatory thinking beyond Europe. Although implementation challenges remain, the EU AI Act currently represents the most comprehensive attempt to translate broad ethical principles into enforceable legal obligations. Its risk-based framework illustrates how governments can balance technological innovation with the protection of fundamental rights through clear regulatory standards, institutional oversight, and meaningful legal accountability. For these reasons, the Act provides an important foundation for the accountability framework proposed in Part IV of this article.

B. The United States: Sector-Specific Regulation and Voluntary Frameworks

The United States has not enacted comprehensive federal AI legislation. Instead, its governance model has evolved through sector-specific rules, executive actions, and voluntary frameworks, reflecting both a longstanding preference for industry-led regulation and the political difficulty of passing broad technology laws. This approach has supported flexibility and innovation but has also produced fragmented oversight and uncertainty regarding legal accountability. The most influential federal initiative to date is NIST's AI Risk Management Framework, released in 2023, [20] which offers voluntary guidance organized around the functions of Govern, Map, Measure, and Manage. Although widely adopted, its nonbinding nature limits consistent compliance and leaves implementation dependent on organizational discretion rather than enforceable obligations.

Executive Order 14110, issued in 2023, [21] directed federal agencies to develop AI governance standards, required developers of advanced AI systems to share safety test results, and instructed the Department of Commerce to establish reporting requirements for high-risk systems.

While significant, its impact is constrained by the inherent instability of executive action, which can be modified or rescinded by subsequent administrations. In contrast, state and local governments have enacted more concrete measures. New York City's Local Law 144 imposes mandatory bias audits for employment algorithms, and states such as California, Colorado, and Illinois have advanced legislation targeting facial recognition, employment screening, and algorithmic insurance practices. This patchwork of state rules creates compliance complexity for national organizations and results in uneven protections for individuals across jurisdictions.

The U.S. model reflects an innovation-first philosophy that prioritizes technological development and addresses harms primarily after they occur. Without comprehensive federal legislation, individuals harmed by algorithmic systems must rely on tort law, civil rights statutes, and consumer protection doctrines that were not designed for AI and generally operate reactively. This stands in contrast to the European Union's preventive, risk-based regulatory approach, which imposes binding obligations and centralized oversight. Each model offers distinct advantages, but the comparison underscores a central point: durable and effective AI governance requires balancing innovation with enforceable legal accountability capable of protecting fundamental rights and sustaining public trust.

C. UNESCO and the International Ethical Framework, OECD AI Principles and Global Coordination

In November 2021, UNESCO's 193 member states adopted the recommendation on the Ethics of AI, [22] the first global standard on AI ethics adopted by a body of that breadth. The Recommendation articulates four core values: respect for human dignity and human rights, fostering just and peaceful societies, ensuring diversity and inclusiveness, and supporting environmental sustainability. From these

values, it derives ten core principles including proportionality and do no harm, safety and security, privacy and data protection, multi-stakeholder governance, responsibility and accountability, transparency and explainability, human oversight, sustainability, awareness and literacy, and fairness and non-discrimination.

The recommendation established the Global AI Ethics and Governance Observatory, designed to serve as a resource for policy makers, regulators, and civil society organizations seeking to translate ethical principles into concrete governance measures. [23] The Observatory's work reflects a recognition that the gap between principle and practice in AI governance is not merely a matter of political will but of institutional capacity, many governments lack the technical expertise and regulatory infrastructure to implement AI governance effectively, and international cooperation is essential to addressing that gap.

The OECD's (Organisation for Economic Co-operation and Development) 2019 AI Principles similarly provide a non-binding but influential framework for trustworthy AI, endorsed by more than fifty countries. [24] They emphasize human-centered values, transparency, robustness, and accountability, and recommend that governments invest in research, strengthen digital ecosystems, and pursue international cooperation. As noted in the document, "the OECD Principles have served as a reference point for national AI strategies and regulatory frameworks worldwide," including the EU AI Act, which reflects the OECD's focus on risk management and human oversight. Together, UNESCO and OECD demonstrate that while voluntary frameworks can establish global consensus on core values, they cannot by themselves ensure consistent, enforceable protection for individuals harmed by AI systems, highlighting the need for jurisdictions to build binding regulatory structures on top of these shared ethical foundations.

Comparative Analysis

Framework	Legal Status	Approach	Enforcement
EU AI Act	Binding law	Risk-based classification	Fines up to 7% global turnover
U.S. AI RMF	Voluntary	Sector-specific guidance	No direct penalties
NYC Local Law 144	Binding law	Domain-specific mandate	Municipal enforcement
UNESCO Recommendation	Non-binding	Principles based	None
OECD AI Principles	Non-binding	Principles based	None

The comparison reveals a consistent pattern. Non-binding frameworks have succeeded in articulating shared values such as transparency, accountability, and human oversight, yet they fall short of providing reliable, enforceable protections for individuals harmed by AI systems. The EU AI Act demonstrates that robust, binding regulation is attainable. The broader challenge for both national and international governance is to build upon that foundation rather than retreat to the limitations of voluntary standards.

IV. TOWARDS AN ENFORCEABLE AI ACCOUNTABILITY FRAMEWORK

A. A New Framework for understanding AI Liability: The Developmental Model

Legal systems usually begin with a straightforward question: who bears responsibility? In the context of artificial intelligence, that question becomes far more complex because AI systems do not behave like

ordinary tools. A hammer performs exactly what its user directs. An AI system, by contrast, learns from its training data, feedback, and the environment in which it operates, and its behavior can evolve in ways its developers did not foresee and its deployers may not fully understand.

A more illuminating analogy, and one with genuine legal utility, is the developmental model of human responsibility. Consider the trajectory of a child from infancy to adulthood. An infant arrives in the world without knowledge, without judgement, and without the capacity of moral reasoning. What the child becomes depends on three things: what it is taught, the environment in which it develops, and the decisions it makes as its capacity for independent judgement matures. No single actor bears complete responsibility for the adult the child becomes. Responsibility is distributed across parents, educators, institutions and ultimately the individual, in proportion to each actor's role in shaping development.

AI systems follow a strikingly similar trajectory. A newly created model knows only what its training data allows it to know. Like an infant learning language through observation and imitation, an AI system absorbs patterns from the data it is given, first reproducing those patterns mechanically, much as a child mimics sounds before grasping their meaning, and only later gaining the ability to generate new outputs. The quality of that learning depends entirely on what the system is taught and who teaches it. As philosophers of science might note, and as Shakespeare suggested in his reflection on the seven ages of man, what endures is not the individual moment but the lasting imprint of what was built, taught, and passed on. [37] AI systems evolve through new versions, but what persists is the underlying architecture of values or the absence of them, embedded by their creators at the foundation. [38]

This developmental model has direct implications for legal liability. Just as we do not hold a child solely responsible for behavior that reflects the environment in which it was raised, we cannot hold any single actor solely responsible for the harms an AI system produces. Liability must be distributed across the chain of development and deployment, allocated in proportion to each actor's contribution to the harm.

B. A Three-Tier Liability Framework

Drawing on the developmental model, this article proposes a three-tier framework for allocating legal responsibility for AI caused harm.

Tier One: Developer Liability: The developer of an AI system occupies the position of the child's first teacher- the actor who shapes the foundational architecture of the system's knowledge and judgement. If a developer trains an AI system on biased, incomplete, or deliberately misleading data, the resulting harms are properly attributed to that foundational decision. Developer liability should attach when the training data was known or reasonably knowable to be biased or defective; when the system's design incorporated insufficient safeguards against foreseeable misuse; or when the developer failed to document known limitations and risks in a manner accessible to downstream deployers.

This standard reflects established principles of products liability law. [25] A manufacturer that releases a defective product into the marketplace is responsible for the harm that defect causes, even when the injury occurs long after and far from the original act of manufacture. AI systems function as products in this legal sense, and their developers occupy the role of manufacturers whose foundational design choices can give rise to downstream harm.

Tier Two: Deployer Liability

The organization that deploys an AI system in a commercial or

governmental context occupies the position of the parent or institution that shapes the environment in which development continues. A company that takes an AI system and integrates it into consequential decision making: hiring, lending, medical diagnosis, criminal risk assessment assumes responsibility for ensuring that the system performs appropriately in that specific context. Deployer liability should attach when the organization failed to conduct adequate pre-deployment testing; when it deployed a system in a domain for which it was not designed or validated; when it failed to implement human oversight mechanisms for high-stakes decisions; or when it continued to use a system after known failures were identified.

The Air Canada chatbot case, [26] discussed in Part II, illustrates this principle. Air Canada could not disclaim responsibility for its chatbot's outputs on the grounds that the chatbot operated autonomously. The Tribunal correctly recognized that an organization cannot benefit from automation while escaping the legal consequences of its automated systems behavior. The deployer who captures the value of AI must also bear the risk.

Tier Three: Shared and Systemic Liability

Some AI-related harms cannot be traced to a single actor; instead, they arise from failures spread across the entire chain of development, deployment, and regulation. When the COMPAS algorithm generated racially biased risk scores that influenced sentencing decisions, responsibility did not rest solely with Equivant, the developer, nor solely with the courts that relied on those scores, nor with the jurisdictions that authorized their use. Each played a part in a system that produced unjust outcomes. A meaningful liability framework must therefore be capable of addressing this kind of distributed causation, where harm results from the combined actions and omissions of multiple actors rather than from one identifiable source.

For harms of this nature, this article advances a shared-responsibility model inspired by the market-share liability doctrine developed in pharmaceutical litigation. [27] When multiple actors across the AI development and deployment chain contribute to a harmful outcome and the specific source of causation cannot be pinpointed, liability should be distributed among them according to their respective roles and the extent to which each failed to meet the applicable standard of care.

C. Mandatory Transparency and Explainability Obligations

Liability systems function best when paired with transparency rules that make the underlying facts discoverable. An injured party cannot pursue a meaningful claim against an AI developer or deployer if the system operates in complete opacity. For that reason, transparency and explainability are not simply ethical preferences they are essential conditions for legal accountability.

The article argues that any AI system used in high-stakes decision-making including employment, credit, healthcare, criminal justice, housing, education, and public benefits, should be subject to mandatory explainability obligations. These obligations should require informing individuals when an automated system influenced a decision affecting them; providing a plain-language explanation of the factors the system considered and how those factors were weighted; documenting the system's training data, known limitations, and error rates; and ensuring a meaningful opportunity to contest automated outcomes before a human decision-maker.

These obligations are entirely within current technical capability. The field of explainable AI already offers developed tools including LIME, SHAP, and counterfactual explanation techniques[28] that can generate clear, human-interpretable accounts of how models reach their

decisions across a wide range of applications. The real barrier is not technological feasibility but the absence of legal requirements. Without legal mandates, organizations have little incentive to prioritize transparency, [39] and commercial pressures will continue to reward opacity rather than accountability.

D. Mandatory Human Oversight in High Stakes Domains

No AI system, however sophisticated, should be permitted to make final, unreviewable decisions in matters affecting fundamental rights and interests without meaningful human oversight. This principle reflected in the EU AI Act's human oversight requirements [29] and in UNESCO's Recommendation should be translated into binding legal obligations in the United States and other jurisdictions that have not yet adopted comprehensive AI regulation.

Human oversight obligations should ensure that in domains such as criminal justice, healthcare, employment, and financial services, AI outputs serve as inputs to human judgment rather than substitutes for it. A COMPAS risk score should guide a judge's reasoning, not replace it. An AI-generated diagnostic suggestion should inform a physician's assessment, not override it. For oversight to be meaningful rather than symbolic, human decision-makers must understand the system's limitations well enough to exercise genuine, independent judgment.

E. Independent Algorithmic Auditing

Transparency and human oversight requirements are most effective when supported by independent verification. This article proposes the establishment of mandatory independent algorithmic auditing requirements for high-risk AI systems, modelled on financial auditing frameworks. [30] Just as publicly traded companies are required to have their financial statements audited by independent certified accountants, organizations deploying high risk AI systems should be required to submit those systems to independent technical audit at regular intervals and before significant updates.

Audits should assess the accuracy and reliability of the system across demographic groups; the quality and representativeness of training data; the effectiveness of bias mitigation measures; and compliance with applicable transparency and explainability requirements. Audit results should be made publicly available, enabling regulators, researchers, civil society organizations, and affected communities to assess AI systems performance and hold deployers accountable.

F. Federal Coordination and the Path Forward

Although recent state and local initiatives demonstrate growing recognition of the need to regulate AI, the resulting patchwork of laws cannot provide the consistency or legal certainty required for effective national governance. Varying regulatory standards increase compliance burdens, create opportunities for regulatory arbitrage, and produce uneven protections for individuals depending on the jurisdiction in which they reside.

V. CONCLUSION

Artificial Intelligence, like every transformative technology before it, will continue to evolve. Systems that seem sophisticated today will be superseded by more capable successors. Algorithms will be retrained, updated, deprecated, and replaced. What will not automatically change unless we deliberately design it to, is the architecture of accountability within which those systems operate. The versions pass. The values, or their absence, persist.

This is the central lesson of the developmental model proposed in Part IV. A child shaped by neglect, bias, or deliberate miseducation does not simply outgrow those influences upon reaching adulthood. The effects compound. They shape decisions, relationships, and institutions long after the original conditions that produced them have changed. Artificial

intelligence systems operate by the same logic. A model trained on biased data, deployed without meaningful oversight, and shielded from legal accountability does not become more just as it scales. It becomes more consequential and the harms it produces become harder to trace, harder to remedy, and harder to reverse.

The governance frameworks surveyed in this article represent genuine progress. The EU AI Act demonstrates that comprehensive, binding AI regulation is legally achievable and politically possible. UNESCO's global ethical framework demonstrates that the core values of human dignity, transparency, and accountability command genuine international consensus. The emerging body of case law, from the Air Canada chatbot ruling to the judicial sanctions imposed on attorneys who submitted AI-generated fabrications demonstrates that existing legal institutions are capable of adapting to the challenges AI presents, even in the absence of specific legislative mandate.

But progress is not sufficiency. Voluntary principles without enforcement mechanisms are statements of aspiration, not instruments of governance. A framework that protects individuals in the European Union but leaves those in other jurisdictions without meaningful recourse is a framework that has chosen convenience over justice. A liability system that diffuses responsibility so thoroughly that no single actor can be held accountable is a system that, in practice, holds no one accountable at all.

The question this article ultimately poses is not technical. It is not whether artificial intelligence can be made safer, more transparent, or more explainable --- the evidence suggests that it can. The question is whether democratic societies will choose to require it. Whether the governments that claim to represent human dignity will build

institutions capable of defending it. Whether the legal systems designed to provide remedy for harm will extend that protection to harms produced by machines as readily as they extend it to harms produced by people.

Shakespeare wrote that all the world is a stage, and the men and women merely players; each making their exits and their entrances, each playing many parts. Artificial intelligence has now joined the cast. It did not write its own lines. It learned them from us, from our data, our decisions, our values, and our failures. What it does with those lines, on the stages that matter most, the courtroom, the hospital, the hiring office, the sentencing hearing will reflect not the intelligence of the machine but the wisdom, or the negligence, of the society that built it.

The systems we design today will outlast the intentions of their designers. The question worth sitting with; the one this article hopes to leave unresolved enough to demand an answer, is this: when the algorithm makes the decision that changes a life, whose values will it be acting on?

And did we ever actually choose them?

Notes

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[3] Regulation (EU) 2024/1689 of the European Parliament and of the Council of 13 June 2024 Laying Down Harmonised Rules on Artificial Intelligence (Artificial Intelligence Act), 2024 O.J. (L 1689) 1.

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[5] New York City Local Law 144 of 2021, Admin. Code § 20-871 (2021).

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[7] *Id.*

[8] Julia Angwin, Jeff Larson, Surya Mattu, and Lauren Kirchner, "Machine Bias," ProPublica, May 23, 2016, <https://www.propublica.org/article/machine-bias-risk-assessments-in-criminal-sentencing>.

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[12] Biometric Information Privacy Act, 740 Ill. Comp. Stat. 14/1 (2008).

[13] *Mata v. Avianca, Inc.*, 678 F. Supp. 3d 443 (S.D.N.Y. 2023).

[14] *Moffatt v. Air Canada*, 2024 BCCRT 149 (B.C. Civil Resolution Tribunal 2024).

[15] Ziad Obermeyer, Brian Powers, Christine Vogeli, and Sendhil Mullainathan, "Dissecting Racial Bias in an Algorithm Used to Manage the Health of Populations," *Science* 366, no. 6464 (2019): 447--453.

[16] Roxana Daneshjou et al., "Disparities in Dermatology AI Performance on a Diverse, Curated Clinical Image Set," *Science Advances* 8, no. 31 (2022), <https://doi.org/10.1126/sciadv.abq6147>.

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[18] Regulation (EU) 2024/1689, *supra* note 3.

[19] Regulation (EU) 2024/1689, art. 99(3).

[20] National Institute of Standards and Technology, *supra* note 4, at 6--8.

[21] Exec. Order No. 14,110, 88 Fed. Reg. 75,191 (Nov. 1, 2023).

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